BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLY PROGRESS REPORT (01/06/2026 – 06/06/2026)

WEEK 3

AUTOMATIC GEO-REFERENCING	**AUTOMATED GCP DETECTION AND GEO-TIFF GENERATION**

PROJECT DESCRIPTION : AUTOMATED GCP DETECTION AND GEO-TIFF GENERATION

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GROUP ID : 2026G262

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COLLEGE NAME : GatiShakti Vishwavidyalaya, Vadodara

01/06/2026	**Created the development environment and installed the core Next.js project framework.** **Configured the base API settings to link the frontend interface with the FastAPI backend server.** **Designed a fixed 3-column workspace grid layout in Dashboard.tsx.**
02/06/2026	**Developed ControlPanel.tsx containing live coordinate and resolution telemetry tracking.** **Integrated Leaflet maps into the center panel to display interactive, scrollable satellite views.** **Built a drag-and-drop zone to let users upload reference images easily with local previews.** **Added quality classifier buttons to pass image properties (sharp/medium/blurry) forward.**
03/06/2026	**Added simple toggle switches to let users flip the image upside down or sideways.** **Made the uploaded image flip instantly on the screen so the user sees the changes in real-time.** **Programmed the page to bundle the image file and all selected settings together so they are ready to send to the server.**
04/06/2026	**Built a dedicated pop-up viewing window (PipelineViewer.tsx) to look inside the processing pipeline.** **Created a sidebar menu that lets users click through the 5 separate stages of image processing.** **Set up the system to receive data from the backend and show text logs and images live as they are finished.**
05/06/2026	**Split the inspection layout into a coordinate data table opposite the visual view workspace.** **Built custom math handlers managing smooth mouse drag-to-pan and scroll wheel-to-zoom actions.** **Authored bounding constraints trapping the translation coordinates within a strict 30% out-of-frame limit.**

06/06/2026	**Constructed a high-density log grid showing point coordinates, drift calculations, and individual error scores.** **Layered a reactive SVG canvas plotting dynamic targeted highlights and explicit error tracking vectors.** **Installed total analytical summary cards (RMSE/MAE) alongside a direct client-side QGIS .points data exporter.**

Next Week Plan	Build an interactive canvas layer for manual GCP refinement to optimize georeferencing accuracy metrics, and containerize the application stack for cloud deployment.

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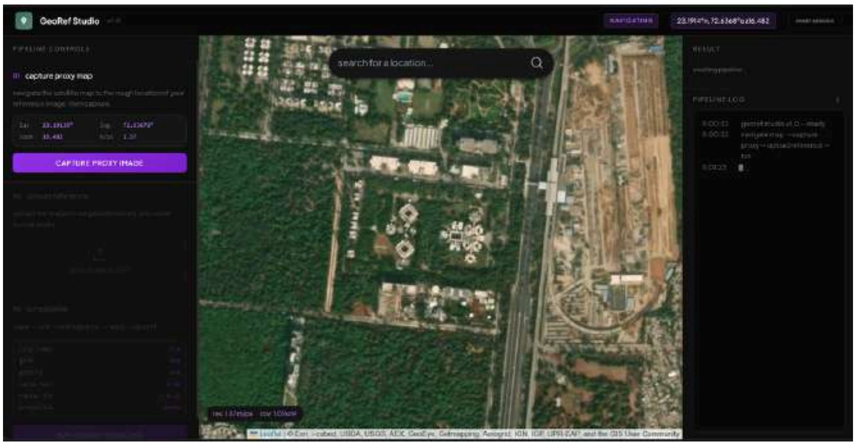

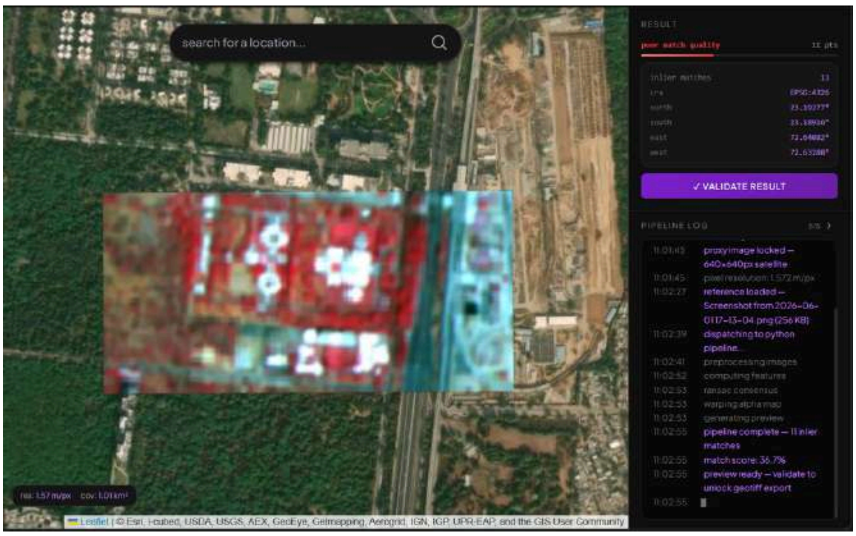

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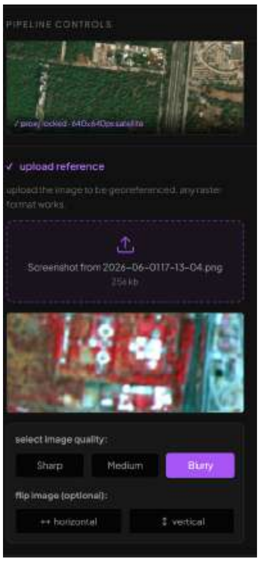

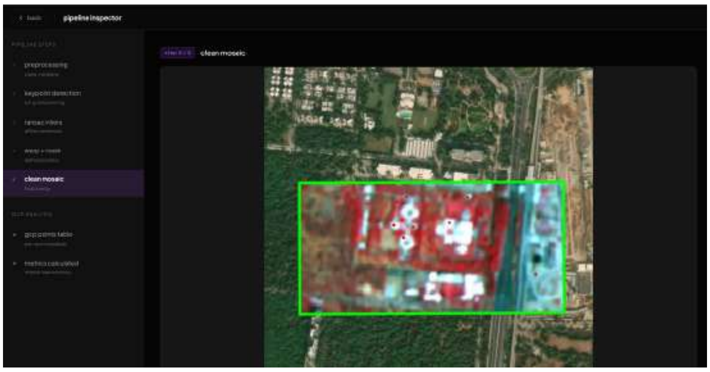

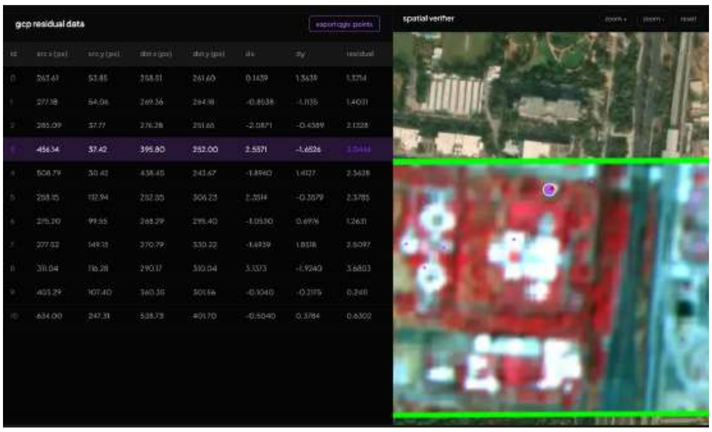

id	src x (px)	src y (px)	dst x (px)	dst y (px)	dx	dy	residual
0	263.61	53.85	258.51	261.60	0.1639	1.3639	1.3714
1	277.18	54.06	269.36	264.18	-0.8538	-1.1135	1.4031
2	285.09	37.77	276.28	251.65	-2.0871	-0.4389	2.1328
3	456.34	37.42	395.80	252.00	2.5571	-1.6526	[illegible]
4	508.79	30.42	438.45	243.67	-1.8940	1.4127	2.3628
5	258.15	132.94	252.05	306.23	2.3514	-0.3679	2.3785
6	275.20	99.55	268.29	295.40	-1.0530	0.6976	1.2631
7	277.02	149.13	270.79	330.22	-1.6939	1.8518	2.5097
8	311.04	116.28	290.17	310.04	3.1373	-1.9240	3.6803
9	403.29	107.40	360.35	301.56	-0.1040	-0.2175	0.2411
10	634.00	247.31	538.73	401.70	-0.5040	0.3784	0.6302

references:

Next.js - https://nextjs.org

Leaflet - https://leafletjs.org

FastAPI - https://fastapi.tiangolo.com

OpenCV - https://opencv.org

Rasterio - https://rasterio.readthedocs.io

QGIS - https://qgis.org